CSE 472: Machine Learning Sessional
Assignment 1
Data Preprocessing & Feature Engineering with a Feed Forward Neural Network Pipeline

Department of Computer Science and Engineering
Bangladesh University of Engineering and Technology

Prepared by: Ananya Shahrin Promi
ID: 2005079

December 17, 2025

Short Report on Architectures

In this report, we summarize the Feed Forward Neural Network (FNN) architectures designed and evaluated for predicting diabetes based on the processed dataset. Three architectures were experimented with. For each architecture, the number of layers, neurons per layer, activation functions, and loss functions are described. Training and validation loss plots are included to justify the choice of the final model.

Architecture Summary

Model	Hidden Layers	Neurons per Layer	Activation Function	Loss Function
Model 1	2	32, 16	ReLU + Sigmoid	BCELoss
Model 2	2	64, 32	ReLU + Sigmoid	BCELoss
Model 3	3	128, 64, 32	ReLU + Sigmoid	BCELoss
Model 4	2	64, 32	Tanh + Sigmoid	BCELoss
Model 5	4	128, 64, 32, 16	ReLU + Sigmoid	BCELoss
Model 6	3	64, 32, 16	Leaky ReLU + Sigmoid	BCELoss

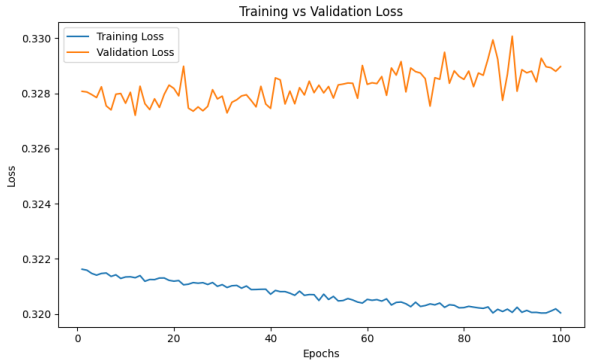
Table 1: Summary of Neural Network Architectures

Performance Metrics

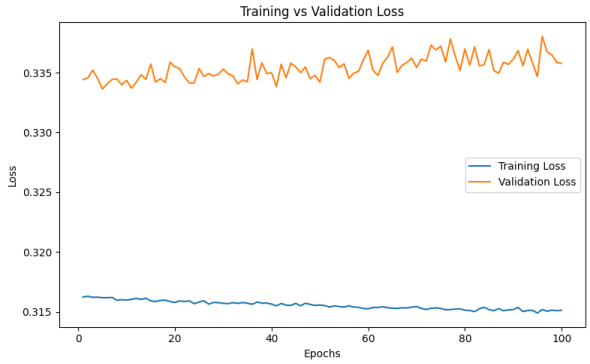
Model	Accuracy	Precision	F1-score	AUROC
Model 1	0.8996	0.2857	0.0015	0.5350
Model 2	0.8994	0.0	0.0	0.5435
Model 3	0.8968	0.1969	0.0184	0.5507
Model 4	0.8995	0.25	0.0031	0.5267
Model 5	0.8978	0.2198	0.0149	0.5489
Model 6	0.8989	0.1944	0.0053	0.5453

Table 2: Performance Metrics of Different FNN Models

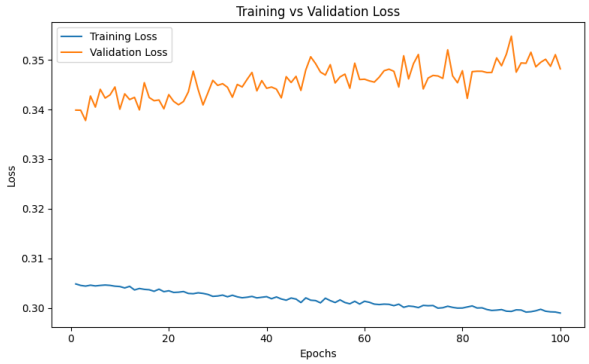
Training vs Validation Loss



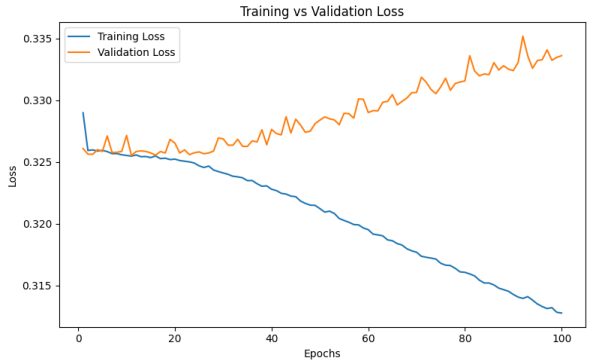
(a) Model 1



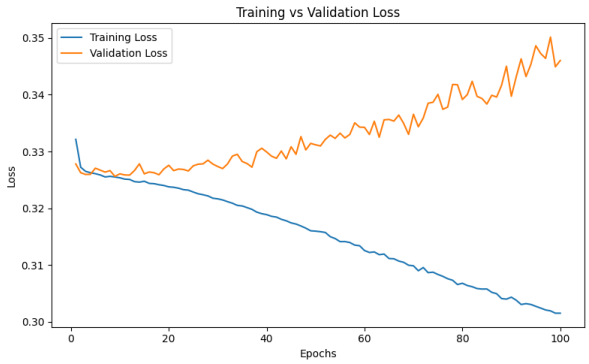
(b) Model 2



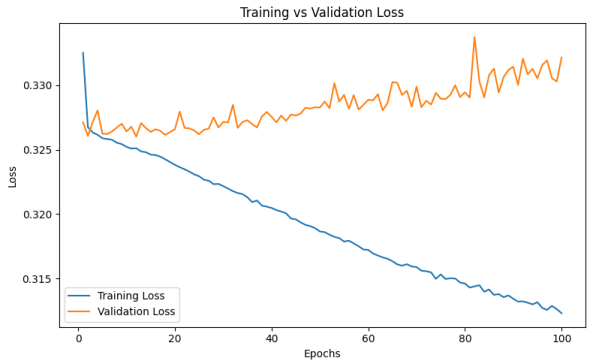
(c) Model 3



(d) Model 4

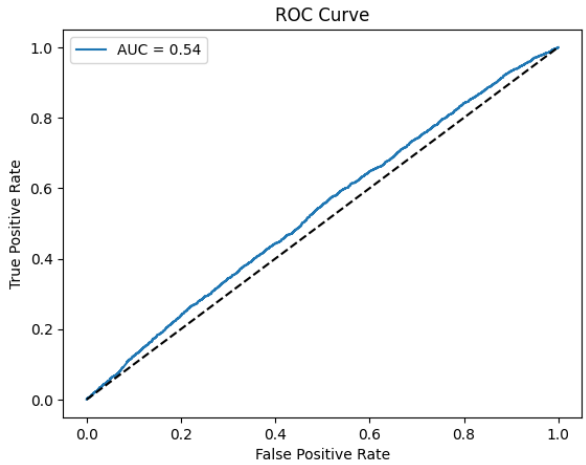


(e) Model 5

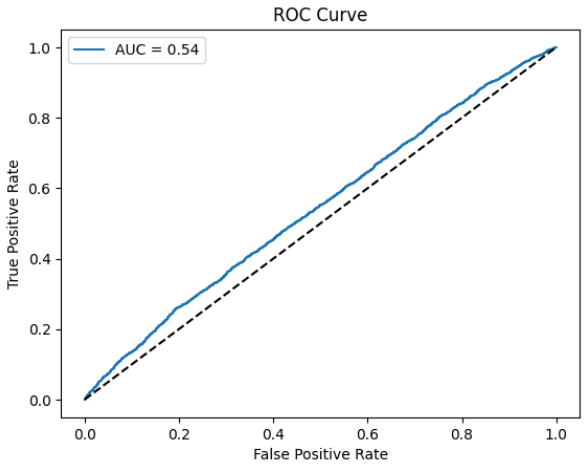


(f) Model 6

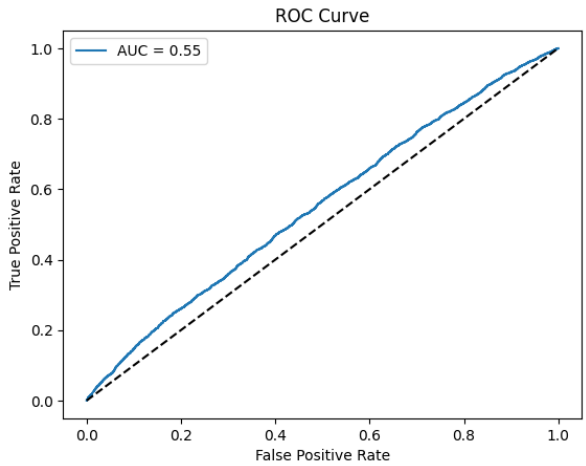
ROC Curve



(a) Model 1



(b) Model 2



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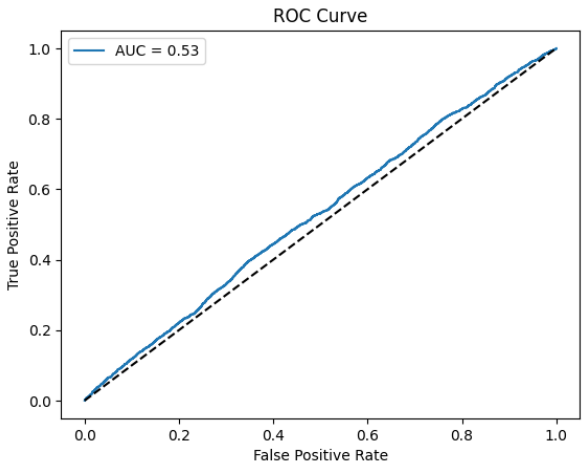
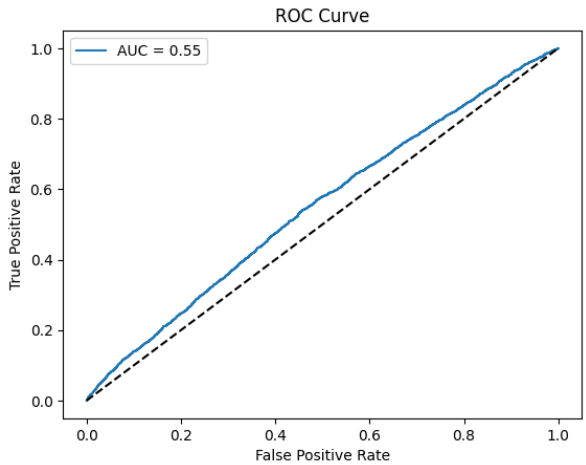
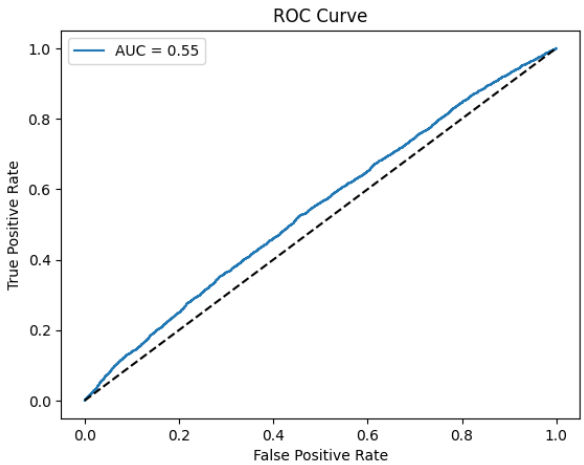


(d) Model 4

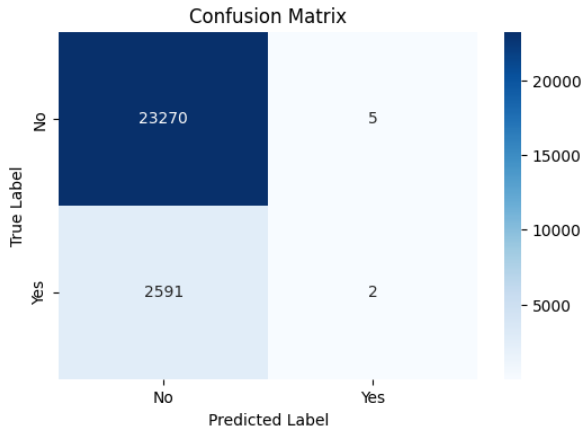


(e) Model 5



(f) Model 6

Confusion Matrix



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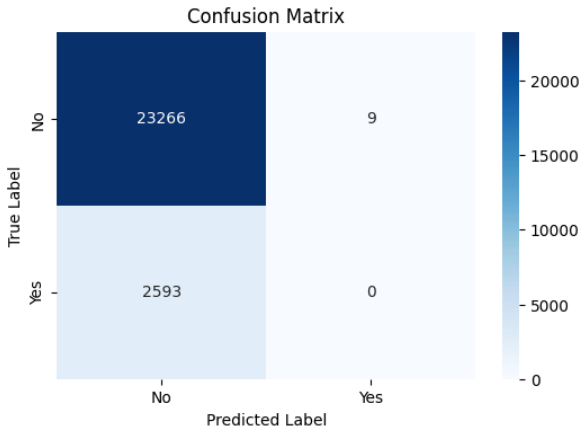
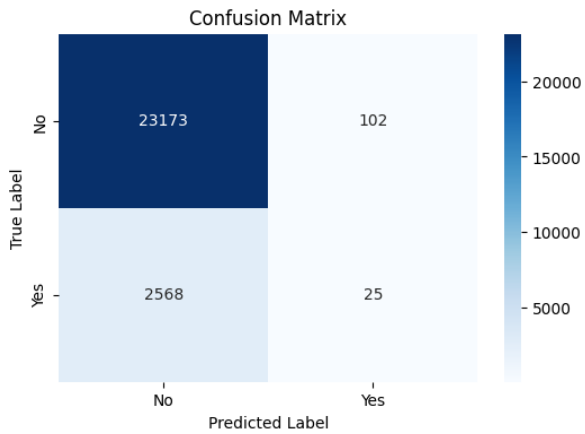
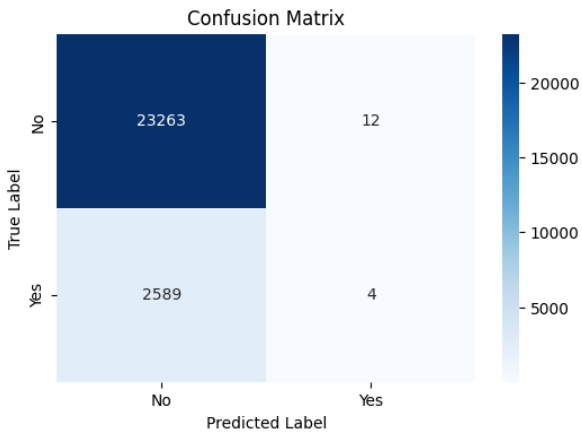


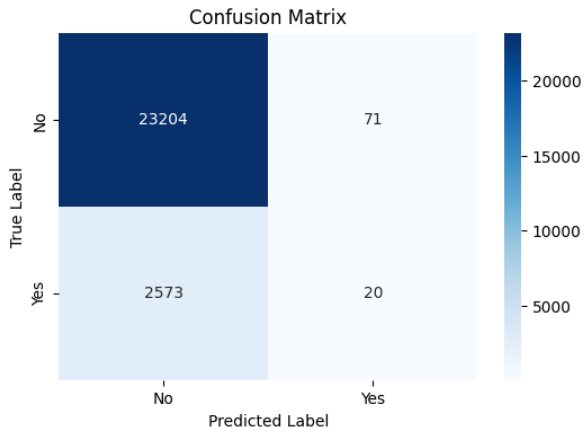
(b) Model 2



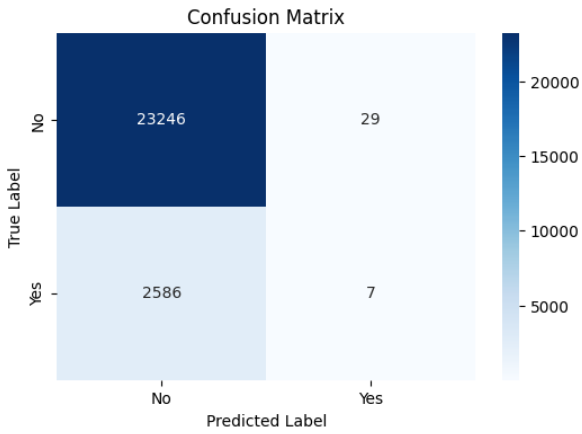
(c) Model 3



(d) Model 4



(e) Model 5



(f) Model 6

Final Model Selection

After evaluating the six feed-forward neural network architectures, Model 3 was chosen as the final model for the diabetes classification task. The decision was based on the following considerations:

- **Discriminative power:** Model 3 achieved the highest AUROC (0.5507), which is the most relevant metric for imbalanced datasets, indicating superior ability to distinguish diabetic from non-diabetic cases.
- **Reasonable accuracy:** Although overall accuracy was comparable across models, Model 3 maintains competitive predictive performance (0.8968) while focusing on class separation.
- **Network architecture:** Model 3 uses three hidden layers, providing sufficient capacity for learning complex relationships without overfitting.
- **Stable training:** Training and validation loss curves for Model 3 demonstrate consistent and smooth convergence, supporting its reliability.

Therefore, Model 3 represents the best trade-off between network complexity and predictive performance for this task.